SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I Rakshana.AR (192126002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ar. R

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1) Cloud Auto-Scaling & Load Balancing During Seasonal Sales:

Cloud auto-scaling automatically increases or decreases computing resources according to application demand. Load balancing distributes incoming user requests across multiple servers to prevent any single server from becoming overloaded.

During seasonal sales, an e-commerce website may experience a sudden increase in visitors. Auto-scaling adds more servers to handle the increased traffic, while load balancing distributes requests among the available servers. This reduces server overload, minimizes downtime, & provides faster page loading & a better user experience.

Monitoring tools continuously track CPU usage, memory, traffic, & response time. When resource usage exceeds predefined limits, they help trigger scaling actions & identify performance issue.

Eg: During a festive sale on Amazon, cloud infrastructure can add computing resources to handle increased customer visits & distribute requests across servers, helping the website remain responsive.

2) Secure Cloud Storage for Backup Data:

Cloud Storage solutions store backup data across multiple storage devices or locations to improve durability & availability. Data redundancy maintains additional copies of information so that data can be recovered if a device or location fails.

Encryption protects sensitive backup data by converting it into an unreadable format without the appropriate decryption key. Access control restricts data access to authorized users through permissions & authentication.

Cloud storage also provides cost advantages because organizations can pay for the storage they use instead of purchasing & maintaining expensive physical storage devices. It reduces hardware, maintenance, & infrastructure costs.

Eg: A hospital can store encrypted patient-record backups in cloud storage. If its local server is damaged by a fire, the hospital can restore the records from cloud backups & continue its operations.

3) Content Delivery Networks (CDNs) for Global Applications.

A Content Delivery Network (CDN) is a geographically distributed network of servers that delivers content from locations closer to users. It reduces latency by minimizing the distance data must travel & decreases the load on the original server.

Edge servers store cached copies of frequently accessed content near users. This improves loading speed, reduces buffering, & provides a smoother experience across different regions, cloud providers support global availability through multiple geographic regions, distributed infrastructure, & redundancy mechanisms.

Eg: A global video-streaming application can use a CDN to deliver videos from nearby edge servers. Users in India & Europe can access content with less delay, while cached content with less delay, while cached content reduces repeated requests to the main server.

4) Cloud Security & Regulatory Compliance in Financial Institutions.

Cloud provides support regulatory compliance by offering security controls, audit logs, compliance reports, & tools for monitoring access to data. Organizations must select services & configurations that meet the applicable financial regulations & security requirements.

Logging records activities such as user logins, transactions, and changes to data. Monitoring helps detect suspicious activities, while ~~det~~ encryption protects financial information & role-based access control ensure that only authorized employees can access sensitive information.

Data sovereignty requirements can be met by selecting appropriate cloud regions & configuring storage & processing locations according to applicable laws & regulations. The organization must regulations.

5) Containerization, Virtual Machines, & orchestration:

Containers package an application with its libraries & dependencies, allowing it to run consistently across different environments. Virtual machines (VMs) in contrast, run complete guest operating systems on virtualized hardware. Containers share the host operating system kernel, so they generally start faster & use fewer resources than VMs.

Container orchestration tools such as Kubernetes automate container deployment, scaling, load distribution, & recovery when containers fail. This reduces manual administration & helps applications handle changing workloads.

Containers are portable because the same container image can run across compatible development systems, on-premises servers, & different cloud platforms. However, compatibility & configuration requirements must still be considered.